

Unifying Theory of Dynamic Aether Mechanics: Bound Core Electron Model

Erik Otto

Working Paper in the Dynamic Aether Mechanics Framework

Abstract

Paper 7 [6] identifies the electron as a quantized vortex ring in the Aether condensate, with mass determined by the Donnelly kinetic energy formula. The vortex core is treated as a depleted region where the condensate density drops to zero at the healing length ξ . This paper proposes a refinement: the Dynamic Casimir Effect (DCE), which operates at the vortex boundary, creates bound Aether that migrates inward under the Bernoulli pressure gradient and accumulates as a toroidal ring in the vortex core. The bound ring grows until the boundary velocity drops below the DCE permanence threshold, at which point only transient binding (gravity) occurs and the electron mass stabilizes. This self-regulating mechanism explains why the electron has the mass it does. The self-consistent vortex equation, corrected for core energy (+5–14%) and the Bernoulli density deficit (−1.7%), predicts $m_e \approx 0.4\text{--}0.6$ MeV across the allowed parameter range and matches the observed 0.511 MeV for $m_A \approx 616\text{--}644$ MeV/ c^2 , well within the independently constrained convergence zone. A companion derivation obtains Coulomb's constant from the ring's spin-phase winding pump ($ke^2 = \rho_A \kappa_e^2 r_{\text{loop}}^2 / 8\pi$) and the Bohr magneton with $g = 2$ exactly from the same geometry, tying the electric, magnetic, and mass sectors to a single topological postulate: one winding advance per Compton cycle.

This working paper explores a specific mechanism within the Dynamic Aether Mechanics framework. It references results from Paper 1 (General Overview) [1], Paper 2 (Special Relativity) [2], Paper 3 (Gravity) [3], Paper 4 (Magnetic and Electric Fields) [4], Paper 5 (Strong and Weak Forces) [5], and Paper 7 (Quantum Mechanics) [6] of the main series. The ideas presented here are under development and may be incorporated into the main papers in revised form.

1. Introduction

Paper 7 [6] establishes that the electron is a quantized vortex ring in the Aether condensate with circulation $\kappa_e = h/(2m_A)$. The electron rest energy is set equal to the Donnelly kinetic energy of the circulatory flow:

$$m_e c^2 = \frac{1}{2} \rho_A \kappa_e^2 r_{\text{loop}} \left[\ln \left(\frac{8 r_{\text{loop}}}{\xi} \right) - \frac{1}{2} \right] \quad (1)$$

where $r_{\text{loop}} = \hbar/(m_e c)$ is the loop radius (reduced Compton wavelength), $\xi = \hbar/(m_A c \sqrt{2})$ is the healing length, and the core is treated as a depleted region where the condensate density drops to zero.

This treatment leaves two questions open:

1. **Where does the DCE product go?** Paper 1 [1] establishes that the spinning vortex core drives the Dynamic Casimir Effect: transient binding produces gravity, and permanent binding (above the DCE threshold) produces mass. But the permanently bound material must go somewhere. The current model is silent on its fate.
2. **Why does the electron have the mass it does?** Eq. (1) determines m_e given m_A and ρ_A , but provides no mechanism for *why* this particular vortex energy is selected from among all possible configurations.

This paper proposes that both questions have the same answer: the DCE permanently binds Aether at the vortex boundary, the bound material migrates inward under the Bernoulli pressure gradient and accumulates as a toroidal ring in the core, the ring grows until the boundary velocity drops below the permanence threshold, and the electron mass stabilizes at this self-regulating equilibrium.

2. Formation Model

The electron forms through the following sequence:

1. **Vortex ring creation.** In the high-energy early universe, turbulent Aether flows create quantized vortex rings with the minimum (spin- $\frac{1}{2}$) circulation quantum $\kappa_e = h/(2m_A)$. Initially these are pure fluid vortices with no bound material—the core is depleted at the healing length as in the standard model.
2. **DCE creates matter.** The circulation velocity at the healing length is (Section 3.1):

$$v(\xi) = \frac{c}{\sqrt{2}} \approx 0.707 c \quad (2)$$

This exceeds the DCE permanence threshold, so the rapidly-circulating boundary continuously converts vacuum fluctuations into permanent bound Aether [8].

3. **Matter migrates inward.** The Bernoulli pressure in the vortex is lowest at the core (Section 3.2). Bound material, no longer part of the superfluid flow, is pushed inward by the pressure gradient and accumulates along the vortex tube axis, forming a toroidal ring of bound Aether tracing the major radius r_{loop} .

4. **Ring grows.** As more bound material falls to the center, the ring expands radially outward from the tube axis. The effective core boundary moves from ξ to a larger radius r_{eq} .
5. **Boundary velocity drops.** The circulation κ_e is topologically quantized and fixed, but the velocity at the boundary is $v(r) = \kappa_e / (2\pi r)$. As r_{eq} increases, the boundary velocity decreases.
6. **Permanence threshold reached.** When $v(r_{\text{eq}})$ drops to v_{perm} —the minimum velocity for permanent DCE binding—the vortex can no longer create permanent mass. Only transient bind–unbind cycling occurs at the boundary, producing the asymmetric pressure oscillation that manifests as gravity [3].
7. **Mass stabilizes.** The electron’s mass is the sum of the remaining circulation kinetic energy and the bound ring’s rest mass at this equilibrium point.

This is a *self-regulating* process: the creation of mass enlarges the core, which reduces the boundary velocity, which eventually shuts off mass creation. The electron mass is not a free parameter but an equilibrium outcome determined by medium properties.

2.1 Energy Conservation

The DCE mechanism converts energy from the vortex’s circulation into bound material—the moving boundary does work on the vacuum fluctuations, and the energy is drawn from the kinetic energy of the flow. As the ring grows, the circulation KE decreases by the same amount that the ring mass increases. The total energy is approximately conserved during the formation process:

$$m_e c^2 \approx E_{\text{KE,initial}} = E_{\text{KE,final}} + m_{\text{ring}} c^2 \quad (3)$$

The electron’s final mass is set by the initial vortex energy at creation, not by the partitioning between KE and ring mass. The circulation quantum $\kappa_e = h / (2m_A)$ is topologically fixed and cannot change; the KE reduction comes from the shrinking integration domain as the core expands from ξ to r_{eq} .

3. Key Equations

3.1 Circulation Velocity

For a quantized vortex with circulation $\kappa_e = h / (2m_A) = \pi \hbar / m_A$, the flow velocity at distance r from the tube axis is:

$$v(r) = \frac{\kappa_e}{2\pi r} \quad (4)$$

At the healing length $\xi = \hbar/(m_A c \sqrt{2})$, the velocity evaluates to $v(\xi) = c/\sqrt{2} \approx 0.707 c$ (Eq. 2). This result is exact and independent of m_A : the circulation quantum and the healing length are both proportional to $1/m_A$, so m_A cancels.

3.2 Bernoulli Pressure

The pressure distribution in the vortex flow:

$$P(r) = P_\infty - \frac{1}{2} \rho_A v^2(r) = P_\infty - \frac{\rho_A \kappa_e^2}{8\pi^2 r^2} \quad (5)$$

Pressure is lowest at the core center and increases outward. Bound material (not participating in superfluid flow) experiences a net inward force from the surrounding higher-pressure fluid, driving inward migration.

3.3 Equilibrium Condition

The bound ring grows until the boundary velocity reaches the permanence threshold:

$$v(r_{\text{eq}}) = v_{\text{perm}} \implies r_{\text{eq}} = \frac{\kappa_e}{2\pi v_{\text{perm}}} \quad (6)$$

For $v_{\text{perm}} < c/\sqrt{2}$, the ring extends beyond the healing length: $r_{\text{eq}} > \xi$.

4. The Electron Mass Equation

4.1 Core Energy

The Donnelly formula [7] (Eq. 1) computes only the kinetic energy of the circulating flow outside the core. In a standard GPE vortex, the total energy also includes the core's gradient and potential energy—the cost of depleting the condensate. This core energy is of order $1/\ln(r_{\text{loop}}/\xi)$ of the kinetic energy, or $\sim 5\text{--}14\%$ for the electron's aspect ratio (the exact value for the logarithmic potential used in this theory has not been computed from first principles).

In the bound-core model, the core region is filled with bound material instead of depleted vacuum. The bound ring's rest mass *replaces* the gradient and potential energy of the depleted core—it is the physical realization of the same core energy, not an additional contribution on top of it.

4.2 Bernoulli Density Deficit

The Donnelly formula assumes constant density ρ_A everywhere. But the vortex circulation itself produces a Bernoulli density deficit. This is a genuine density change (not just a pressure change with constant ρ_A) because the vortex circulation is *driven* flow, not free-fall—the persistent spin energy maintains the flow against Bernoulli pressure, creating a venturi-like deficit. This is physically distinct from gravitational inflow, where free-fall dynamics ensure $\Delta\rho = 0$ exactly [2]:

$$\frac{\delta\rho(r)}{\rho_A} \approx -\frac{v^2(r)}{2c^2} = -\frac{\kappa_e^2}{8\pi^2 c^2 r^2} \quad (7)$$

At the healing length ($v = c/\sqrt{2}$), the deficit reaches $\delta\rho/\rho_A \approx -25\%$, dropping rapidly with distance:

r/ξ	v/c	$\delta\rho/\rho_A$
1.0	0.707	-25%
1.5	0.471	-11%
2.0	0.354	-6%
5.0	0.141	-1%

The density deficit reduces the kinetic energy. Integrating the correction $\delta E = \int \frac{1}{2} (-\delta\rho) v^2 dV$:

$$\frac{\delta E_{\text{KE}}}{E_{\text{KE}}} \approx -\frac{1}{8 \ln(r_{\text{loop}}/\xi)} \approx -1.7\% \quad (8)$$

The factor $1/8$ arises because $\kappa_e^2/(16\pi^2 c^2 \xi^2) = 1/8$ for $v(\xi) = c/\sqrt{2}$. For the logarithmic equation of state, the exact compressible Bernoulli gives $\rho(r) = \rho_A e^{-v^2/(2c^2)}$ rather than the linearized $\rho_A [1 - v^2/(2c^2)]$; the integrated KE correction differs by less than 3% between exact and linear forms, so the linearized result is adequate.

The Donnelly formula itself is non-relativistic. In the Aether framework, relativistic effects emerge from the medium at $O(v^2/c^2)$: time dilation from wave path geometry, length contraction from density changes [2]. The Bernoulli density correction is the leading such correction to the vortex energy; higher-order terms are $O(v^4/c^4) \sim 0.4\%$, negligible at this level of precision.

4.3 Combined Mass Equation

Combining all contributions:

$$m_e c^2 \approx \underbrace{\frac{1}{2} \rho_A \kappa_e^2 r_{\text{loop}} \left[\ln \left(\frac{8 r_{\text{loop}}}{r_{\text{eq}}} \right) - \frac{1}{2} \right]}_{E_{\text{KE}} \text{ (circulation)}} \times \underbrace{(1 - 0.017)}_{\text{Bernoulli}} + \underbrace{E_{\text{core}}}_{\sim 5-14\% \text{ of } E_{\text{KE}}} \quad (9)$$

The net correction from the bare Donnelly formula is:

$$m_e c^2 \approx E_{\text{KE}}^{\text{Donnelly}} \times (1 - 0.017 + f_{\text{core}}) \quad (10)$$

where $f_{\text{core}} \approx 0.05-0.14$ is the core energy fraction (model-dependent). The net correction ranges from +3% to +12% above the bare Donnelly value. The loop radius is $r_{\text{loop}} = \hbar/(m_e c)$ (the reduced Compton wavelength), following Paper 7 [6].

5. Mass Predictions

5.1 Donnelly Kinetic Energy

Using the self-consistent condition $r_{\text{loop}} = \hbar/(m_e c)$ and setting $r_{\text{eq}} = \xi$, the Donnelly formula gives:

$$m_e^2 = \frac{\pi^2 \rho_A \hbar^3}{2 m_A^2 c^3} \left[\ln \left(\frac{8 \sqrt{2} m_A}{m_e} \right) - \frac{1}{2} \right] \quad (11)$$

This transcendental equation is solved iteratively. Results for $\rho_A = 5 \times 10^{11} \text{ kg/m}^3$:

m_A (MeV/ c^2)	ξ (fm)	r_{loop} (fm)	m_e (KE only) (MeV)	vs. observed
500	0.28	325	0.61	+19%
550	0.25	354	0.56	+9%
600	0.23	383	0.515	+1%
650	0.21	411	0.48	-6%
700	0.20	439	0.45	-12%
800	0.17	495	0.40	-22%

The bare Donnelly formula gives $m_e \approx 0.4-0.6 \text{ MeV}$ across the full convergence zone (500–850 MeV/ c^2 from Paper 7 [6]), bracketing the observed 0.511 MeV.

5.2 Corrected Prediction

Since DCE converts circulation KE into ring mass (energy conserved, Eq. 3), the total electron mass is set by the initial vortex energy. The Donnelly formula gives only the KE portion. Including the core energy and Bernoulli corrections (Eq. 10), the corrected prediction depends on the core energy fraction f_{core} .

One propagation subtlety must be handled carefully. The energy relation $m_e c^2 = E_{\text{KE}} \times (1 + \text{net})$ does not shift the self-consistent mass linearly, because $E_{\text{KE}} \propto r_{\text{loop}} \propto 1/m_e$: substituting $r_{\text{loop}} = \hbar/(m_e c)$ places the correction in the *squared* mass, $m_e^2 \propto (1 + \text{net})$, so the self-consistent solution shifts by $\sqrt{1 + \text{net}}$ —half the naive exponent:

f_{core}	Net correction	Best-fit m_A	KE only (MeV)	Corrected (MeV)
5%	+3%	616 MeV	0.503	0.511
8%	+6%	625 MeV	0.496	0.511
11%	+9%	635 MeV	0.490	0.511
14%	+12%	644 MeV	0.484	0.511

For any core energy in the 5–14% range, there exists an m_A in the convergence zone that matches the observed mass:

$$\boxed{m_e \approx 0.511 \text{ MeV} \quad (\text{predicted, } m_A \approx 616\text{--}644 \text{ MeV}) \quad \text{vs.} \quad m_e = 0.511 \text{ MeV} \quad (\text{observed})} \quad (12)$$

The Bernoulli density deficit (−1.7%) partially offsets the core energy (+5–14%), and the remaining net correction (+3–12%)—entering as $\sqrt{1 + \text{net}}$ —is absorbed by m_A , which shifts from 600 MeV (bare Donnelly) to 616–644 MeV (corrected).

6. Physical Structure at Equilibrium

6.1 Compression Ratio

Even if the entire electron mass were packed into the core ring (the extreme $f = 1$ case), the compression ratio would be remarkably low:

$$X = \frac{\rho_{\text{core}}}{\rho_A} = \frac{m_e}{\rho_A \cdot 2\pi^2 r_{\text{loop}} \xi^2} \approx 4\text{--}5 \quad (13)$$

(varying from ~ 4.4 at $m_A = 600$ to ~ 4.9 at $m_A = 630$). Only ~ 5 times ambient density even in the extreme case—far below the $\sim 7 \times 10^7$ compression in QCD bridges [5]. This reflects the electron’s small mass (0.511 MeV) spread over a large circumference ($2\pi r_{\text{loop}} \approx 2400 \text{ fm}$).

For the core energy range $f_{\text{core}} = 5\text{--}14\%$, the ring carries only 0.03–0.07 MeV. Spread over the core volume ($\sim 300\text{--}400 \text{ fm}^3$), the ring density is well below ρ_A —the core is sparsely filled, not compressed.

6.2 Self-Regulation

The self-regulation mechanism proceeds as follows:

1. DCE creates bound material at the core boundary where $v > v_{\text{perm}}$
2. Material falls inward, expanding the effective core from ξ to r_{eq}
3. At r_{eq} , the velocity has dropped: $v(r_{\text{eq}}) < v(\xi)$
4. When $v(r_{\text{eq}}) = v_{\text{perm}}$, permanent binding ceases
5. Transient binding continues (producing gravity), but no new mass accumulates

The quantized circulation $\kappa_e = h/(2m_A)$ is topologically fixed—it cannot change. Only the boundary radius changes as the ring grows, reducing the velocity through the $v \propto 1/r$ dependence.

Under energy conservation (Eq. 3), the total electron mass is fixed by the initial vortex energy at creation. The threshold velocity v_{perm} determines only the partitioning between circulation KE and ring rest mass, not the total. The ring mass at equilibrium is:

$$m_{\text{ring}} c^2 = f_{\text{core}} \times m_e c^2 \quad (14)$$

where $f_{\text{core}} \approx 5\text{--}14\%$. The electron remains circulation-dominated at equilibrium.

6.3 The DCE Permanence Threshold

The permanence threshold v_{perm} determines the equilibrium core size $r_{\text{eq}} = \kappa_e/(2\pi v_{\text{perm}})$, the KE/ring mass partitioning, and the detailed structure of the electron. For v_{perm} near $c/\sqrt{2}$ (modest ring growth), the total mass varies only weakly with v_{perm} . A natural estimate from the Schwinger pair-production analogy gives:

$$v_{\text{perm}} \sim \frac{c}{\sqrt{\pi}} \approx 0.56 c \quad (15)$$

This corresponds to $r_{\text{eq}} \approx 1.26 \xi$ —a modest expansion of the core beyond the healing length. Quantifying v_{perm} from first principles is the key open calculation (Section 12).

7. Significance

7.1 Electron Mass from Medium Properties

The electron mass is determined by m_A , ρ_A , and c —all properties of the Aether medium—through a specific transcendental equation (Eq. 11) with no adjustable structure. The mass arises from three interlocking constraints:

- **Circulation quantization:** $\kappa_e = h/(2m_A)$ fixes the flow speed at each radius
- **Quantum minimum size:** $r_{\text{loop}} = \hbar/(m_e c)$ identifies the loop radius with the reduced Compton wavelength (Paper 7 [6])
- **Self-consistency:** m_e must equal the energy of a vortex ring of size r_{loop} , which itself depends on m_e

For $m_A \approx 616\text{--}644$ MeV (within the convergence zone), the corrected prediction matches the observed 0.511 MeV. The inputs m_A and ρ_A are independently constrained by QCD bridge physics [5] and the healing-length convergence [6]. The core energy fraction f_{core} is the remaining model-dependent quantity; computing the vortex core structure in the logarithmic potential would pin down both f_{core} and the best-fit m_A simultaneously.

7.2 Self-Regulating Mass

The bound-core model explains *why* the electron has the mass it does: the DCE mechanism creates mass until it shuts itself off. Every electron reaches the same equilibrium because every minimum-circulation vortex ring has the same κ_e , the same velocity profile, and therefore the same threshold crossing point. Mass universality follows from circulation quantization.

7.3 Open Tube / Closed Tube Taxonomy

The bound electron core ring is a *closed tube* of gently compressed Aether, stabilized by the encircling vortex circulation. The QCD bridge is an *open tube* of strongly compressed Aether, stabilized by quark vortex endpoints [5]. Both are topologically-stabilized bound Aether structures in the same medium, differing in topology (closed vs. open), compression ($\ll 1$ to ~ 5 vs. $\sim 7 \times 10^7$), and energy scale (0.5 MeV vs. ~ 1 GeV/fm).

7.4 What Is Derived vs. Assumed

The self-consistent mass equation (Eq. 11) is derived from the Donnelly vortex energy formula [7], circulation quantization ($\kappa_e = h/(2m_A)$), and the identification $r_{\text{loop}} = \hbar/(m_e c)$ from Paper 7 [6]. The Bernoulli correction (Eq. 8) is derived from the density-pressure coupling in the unbound superfluid. The formation narrative and energy conservation (Eq. 3) are derived from the DCE mechanism and Bernoulli pressure gradient.

Three quantities remain undetermined from first principles: m_A (constrained to 500–850 MeV/ c^2 by QCD bridge physics and healing-length convergence, but not uniquely determined), f_{core} (the core energy fraction, which requires solving the GPE with the logarithmic potential), and $r_{\text{loop}} = \hbar/(m_e c)$ (the loop radius, identified with the Compton wavelength in Paper 7 but not derived from vortex structure). Deriving r_{loop} from first principles remains an open problem (Section 12). Independently determining m_A or f_{core} would reduce the remaining freedom.

8. Relationship to Paper 7

This working paper proposes three modifications to the electron model in Paper 7:

1. **Core interpretation.** The vortex core is filled with bound Aether rather than being a depleted vacuum. The condensate density still drops to zero at $\sim \xi$; the core region contains non-condensate material (bound Aether) rather than nothing.
2. **Energy corrections.** The Donnelly formula undercounts the total energy by $\sim 5\text{--}14\%$ (core energy, model-dependent) and overcounts by $\sim 1.7\%$ (Bernoulli density deficit). The net correction, entering the self-consistent mass as $\sqrt{1 + \text{net}}$, shifts the best-fit m_A from ~ 600 MeV to $\sim 616\text{--}644$ MeV, still within the convergence zone.
3. **Mass origin.** The electron mass is not merely *determined* by the vortex equation—it is *selected* by the self-regulating DCE mechanism. The formation narrative provides a physical reason why the vortex settles at this particular energy.

No equations in Paper 7 need to be changed. The Donnelly formula remains the dominant contribution ($\sim 85\text{--}95\%$ of the total mass). The corrections refine the numerical prediction and shift the optimal m_A within the existing allowed range.

9. The Form Factor Challenge

Electron scattering experiments (Bhabha scattering at LEP, Møller scattering, $g-2$ measurements) find no deviation from the QED point-particle prediction up to momentum transfers $Q \sim 200$ GeV,

corresponding to a spatial resolution of $\sim 10^{-18}$ m [9]. Yet the vortex model predicts an electron with loop radius $r_{\text{loop}} \approx 386 \text{ fm} = 3.9 \times 10^{-13} \text{ m}$ —five orders of magnitude larger. This section examines what the experiments actually constrain and how the vortex model may be consistent.

9.1 What the Experiments Measure

These experiments do not image the electron. They measure scattering cross-sections and compare them to QED predictions for a structureless point particle. “Point-like to 10^{-18} m” means: *the electron’s electromagnetic interactions match QED predictions at all tested momentum transfers*. Any internal structure would modify the electron–photon vertex, producing a form factor $F(Q^2) \neq 1$ that changes the cross-section at the scale $Q \sim \hbar c/r$, where r is the charge radius.

For comparison, the proton has a charge radius of 0.87 fm and its form factor deviates from 1 at $Q \sim 200 \text{ MeV}$, exactly as expected from its spatial extent. The proton’s charge radius (0.87 fm) exceeds its Compton wavelength (0.21 fm) by a factor of 4, so structure is always resolvable.

For the electron vortex, if the charge were distributed around the 386 fm ring, $F(Q^2)$ would deviate from 1 at $Q \sim 0.5 \text{ MeV}$. This is six orders of magnitude below LEP’s reach—the deviation would be enormous and unmissable. The absence of any deviation is a genuine constraint.

9.2 Topological Charge: Confirmed Resolution

The resolution rests on the nature of the electron’s charge in the Aether model. The revised Paper 4 [4] identifies charge as the relative rotation coupling between the bound core ring and its surrounding vortex—co-rotating for the electron ($q = -e$), counter-rotating for the positron ($q = +e$). This coupling is purely topological: a discrete relative winding sign with no continuous deformation between the two configurations and no spatial distribution of charge density.

Charge is therefore a property of the topological defect, not of the 386 fm ring geometry. The ring carries mass (circulation KE) and spin (angular momentum) distributed over its circumference, but the charge itself has no spatial extent. The form-factor prediction is $F(Q^2) = 1$ at all Q , consistent with electron-scattering measurements up to $Q \sim 200 \text{ GeV}$.

The distinction is analogous to that of a magnetic monopole: a monopole’s charge is topological (from the field configuration’s winding number), not geometric (from a spatial charge distribution). Its form factor is $F(Q^2) = 1$ because there is no charge distribution to resolve—the charge *is* the topology. The same applies here: there is no “charge distribution” on the electron ring to resolve, because charge is sourced by the relative winding of ring and vortex, not by anything distributed around the ring itself.

Paper 4’s description of the electric field as spherically symmetric is therefore exact (not approximate): the symmetry arises from topology, and the coherent spin-phase wave field radiated by the ring

inherits this exact spherical symmetry at the monopole level.

9.3 Status

The topological-versus-geometric question that was previously open is answered by the revised Paper 4 [4]: charge is topological, the form factor satisfies $F(Q^2) = 1$ exactly, and the vortex-ring model is consistent with electron-scattering data at all tested momentum transfers. The quantitative Coulomb-constant derivation is developed in the following section (Section 10), where the same relative winding that carries the charge acts as a quantized spin-phase pump; that derivation is independent of—and does not affect—the topological resolution of the form-factor challenge.

10. Derivation of Coulomb’s Constant from the Ring’s Spin-Phase Winding Pump

Paper 4 [4] develops the electric field as a phase-locked interaction between two coherent sources of discrete topological sign, deriving the Coulomb-force *structure* ($1/r^2$ falloff, sign rule) without committing to a specific source realization. This section supplies the quantitative source from the bound core ring topology of Sections 2–9 and derives k . An earlier version of this section modeled the source as an axially oscillating piston radiating density (sound) waves; that route is closed by the channel-selection result below, and by Paper 7’s identification of photons as spin-wave excitations of the spinor condensate [6], which places the electromagnetic sector in the spin channel from the outset.

10.1 Channel Selection: The Source Cannot Be a Density Monopole

For any closed surface around the particle, the monopole (isotropic, $1/r^2$ -force-producing) component of a radiated field is fixed by conservation: the net flux equals the rate of change of the corresponding *content* enclosed. For density (sound) waves the content is displaced volume, and for a structure made of flow the available volume is pinned by an identity: the Bernoulli deficit volume of a flow structure equals its flow energy divided by the bulk modulus, $V_{\text{def}} = E_{\text{flow}}/K_A$. Matching the observed Coulomb force through the acoustic (secondary Bjerknes) route [10] requires a volume amplitude $V_0 = \sqrt{8\pi k e^2 / (\rho_A \omega^2)} \approx 1.4 \times 10^{-40} \text{ m}^3$, whose energy cost is

$$K_A V_0 \approx (75\text{--}110) m_e c^2 \quad (16)$$

—two orders of magnitude beyond the electron’s entire energy budget, which the mass equation of Section 4.3 has already spent on the circulation. (Motion-based sources fare no better: a translating or flexing structure that conserves its volume radiates as a dipole at every instant, whatever its

stiffness.) The density channel therefore cannot host the Coulomb source. The spin channel can: the transverse spin $S_+ = \psi_\uparrow^* \psi_\downarrow$ is a complex field whose phase is *compact*—a precessing source pumps it indefinitely without accumulating content—so monopole sources of spin phase are not content-bounded. The density no-go thus acts as a channel-selection theorem, and it doubles as a structural reason why the density-channel residual (gravity, Paper 3 [3]) is so much weaker than electromagnetism.

10.2 Spin-Phase Stiffness of the Condensate

Write the spinor condensate as $\Psi = (\psi_\uparrow, \psi_\downarrow)$ with $\psi_i = \sqrt{n_i} e^{i\theta_i}$ over a balanced background ($n_\uparrow = n_\downarrow = n/2$, $n = \rho_A/m_A$), and define the total phase $\theta = (\theta_\uparrow + \theta_\downarrow)/2$ and the relative (spin) phase $\varphi = \theta_\uparrow - \theta_\downarrow$. The Gross–Pitaevskii kinetic energy splits exactly:

$$\frac{\hbar^2}{2m_A} [n_\uparrow (\nabla\theta_\uparrow)^2 + n_\downarrow (\nabla\theta_\downarrow)^2] = \frac{K_\theta}{2} (\nabla\theta)^2 + \frac{K_\varphi}{2} (\nabla\varphi)^2, \quad K_\varphi = \frac{\rho_A \hbar^2}{4m_A^2} = \frac{\rho_A \kappa_e^2}{4\pi^2}. \quad (17)$$

The spin-phase stiffness K_φ is purely kinetic—the interaction terms set wave speeds and healing lengths but do not enter Eq. (17)—so the static results below carry no new parameters, and the factor of 4 that previously had to be supplied by the half-quantum assumption emerges here from the two-component kinematics ($\theta_i = \theta \pm \varphi/2$).

10.3 Flux Sources and the Interaction

A particle that pumps relative phase is a flux source of the φ field, with source strength $q = \oint \nabla\varphi \cdot d\mathbf{A}$; by two-component continuity, $q = (2m_A/\hbar n) d(\Delta N)/dt$, where ΔN is the enclosed spin-population imbalance. The energy functional (17) is mathematically identical to incompressible potential flow under $\rho \leftrightarrow K_\varphi$ and volume flux $\leftrightarrow q$, so potential theory gives the interaction of two sources directly: $E_{\text{int}} = K_\varphi q_1 q_2 / (4\pi d)$, and for two synchronized oscillating pumps $q_i(t) = q_0 \sin(\pm\omega t)$ the time-averaged force is

$$F = \frac{K_\varphi q_1 q_2}{8\pi d^2}. \quad (18)$$

The sign structure is that of flux sources of a gradient field (the vortex-gas rule): like signs repel, opposite signs attract. The sign of each pump is its winding sense—co-rotating versus counter-rotating core, exactly the topological charge of Section 9—so Eq. (18) is electrostatics with Gauss-law charges: field lines are steady spin-current streamlines, and Gauss’s law is pump conservation.

10.4 The Winding Pump

The source amplitude follows from the ring's two rotations (§11.2). The bound core co-rotates with the host flow that created it, so the winding rotates poloidally—about the core tube's axis—at the inherited rate

$$\Omega = \frac{v(\xi)}{\xi} = \frac{m_A c^2}{\hbar}, \quad (19)$$

a property of the medium ($v(\xi) = c/\sqrt{2}$ and ξ scale so that m_A cancels from the construction), and hence common to every singly-wound structure—which is what allows unequal-mass charges to remain synchronized. A rigidly rotating pattern changes no enclosed totals; what pumps is the rotation's effect at the boundary: the $m = 1$ winding rotating at Ω makes the *local* relative phase at every boundary point precess uniformly at Ω (an axisymmetric, monopole-compatible drive), which is an AC Josephson condition with bias $\hbar\Omega = m_A c^2$ —one constituent rest energy, the DCE bind-unbind scale of Section 6.3. The boundary layer therefore converts spin population sinusoidally at Ω . The pump amplitude is normalized by the same topological postulate that fixes the effective current in §11: *one winding quantum per cycle*, giving

$$q_0 = 2\pi r_{\text{loop}}, \quad (20)$$

the ring's own circumference. Three computed checks support this normalization without yet deriving it: the required conversion current is $I = n \kappa_e r_{\text{loop}}$ (frequency-independent); the boundary layer's required duty is $2/\pi$ of its population per turn (order unity—compare the factor ~ 300 by which the density channel falls short); and the layer then operates at $\approx 1/(2\pi)$ of its critical spin current, i.e. well subcritical, which is what makes the pumping sinusoidal. The remaining first-principles gap is the boundary-layer conversion calculation itself (Section 12).

10.5 The Coulomb Constant

Combining Eqs. (17), (18), and (20), the frequency, the amplitude conventions, and all effective areas cancel, leaving

$$k e^2 = \frac{K_\varphi q_0^2}{8\pi} = \frac{\rho_A \kappa_e^2 r_{\text{loop}}^2}{8\pi} \quad (21)$$

—Coulomb's constant from the medium density, the circulation quantum, and the loop radius alone. Eliminating $\rho_A \kappa_e^2 r_{\text{loop}}^2$ with the mass equation (Section 4.3) and using $m_e c^2 r_{\text{loop}} = \hbar c$ gives the dimensionless content of the joint model:

$$\alpha^{-1} = 4\pi \left[\ln \left(\frac{8\sqrt{2} m_A}{m_e} \right) - \frac{1}{2} \right] (1 + f_{\text{core}} - 0.017), \quad (22)$$

in which ρ_A cancels entirely and m_A enters only logarithmically. Numerically, Eq. (21) reproduces the observed $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$ to 0.2% at $\rho_A = 5 \times 10^{11} \text{ kg/m}^3$ (the reference density of Section 5.1) and $m_A = 666 \text{ MeV}/c^2$. The substantive tests are not this evaluation, which selects the point, but the joint consistency of Section 10.6 and the parameter-free magneton match of §11, which shares the normalization postulate of Eq. (20).

10.6 Joint Consistency with the Electron Mass

Both the Coulomb constant and the electron mass pin the same combination of medium parameters. Eq. (21) fixes

$$K_C \equiv \frac{\rho_A}{m_A^2} = \frac{8 m_e^2 c^2 e^2 k}{\pi \hbar^4} \approx 3.54 \times 10^{65} \text{ kg}^{-1} \text{ m}^{-3}, \quad (23)$$

while the mass equation fixes ρ_A/m_A^2 up to the core-energy correction. Requiring both—equivalently, solving Eq. (22) for the observed α —demands

$$f_{\text{core}} \approx 0.18\text{--}0.26 \quad (0.22 \text{ at } m_A = 666 \text{ MeV}/c^2), \quad (24)$$

somewhat above the a-priori estimate $f_{\text{core}} \sim 5\text{--}14\%$ of Section 4.1. With that a-priori band the predicted k runs 7–17% high across the convergence zone; exact agreement is a sharp, pre-registered requirement on the GPE core-energy calculation. The joint consistency is a *requirement on* f_{core} , not an independent overdetermination—but it is now stated at a single common density, and the exact-agreement point ($m_A \approx 666 \text{ MeV}/c^2$ at $\rho_A \approx 5.0 \times 10^{11} \text{ kg/m}^3$) coincides with the reference density of the mass analysis rather than sitting at a different one. The requirement is also sensitive to the vortex-energy bracket constant inherited from Paper 7 (the $-\frac{1}{2}$ in Eq. 1); verifying that constant against the vortex-ring literature shifts Eq. (24) accordingly and is flagged in Section 10.8.

10.7 Parameter Constraints

Eq. (23) collapses the two-dimensional (ρ_A, m_A) space to a curve. Along it:

$m_A \text{ (MeV}/c^2)$	$\rho_A \text{ (kg/m}^3)$	$\xi \text{ (fm)}$	$K_A \text{ (J/m}^3)$
500	2.8×10^{11}	0.28	2.5×10^{28}
666	5.0×10^{11}	0.21	4.5×10^{28}
850	8.1×10^{11}	0.16	7.3×10^{28}

The combination $\xi^2 K_A = \hbar^2 K_C / 2 \approx 1.97 \times 10^{-3} \text{ J/m}$ is exactly invariant along the curve. Evaluating the compression ratio of Paper 7 with the lattice-QCD string tension $\sigma \approx 1.6 \times 10^5 \text{ J/m}$ gives

$X_{\text{Coulomb}} = 8\sigma/(\pi\hbar^2 K_C) \approx 1.0 \times 10^8$ ($\log_{10} X \approx 8.0$), about 50% above Paper 7's electron-mass-only estimate $X \approx 7 \times 10^7$, with bridge sound speed $c_s = c/\sqrt{X} \approx 2.9 \times 10^4$ m/s; the shift does not overturn any qualitative strong-force conclusion. Final pinning of the point on the curve awaits the f_{core} calculation (Eq. 24).

10.8 Honest Flags

The derivation rests on stated, checkable assumptions:

1. **Pump normalization (one winding per cycle).** Eq. (20) is the same postulate that yields the Bohr magneton exactly (§11), and the capacity and operating-point checks of §10.4 pass, but the boundary-layer conversion (duty $2/\pi$ per turn) has not been derived from the GPE.
2. **Species inheritance of the charge quantum.** Eq. (20) uses the *electron's* loop radius. Because m_e is itself fixed by the medium (Section 4.3), $q_0 = 2\pi\hbar/(m_e c)$ can be read as a medium constant—the charge quantum of the condensate's ground ring mode—but the mechanism by which a muon or proton inherits exactly this quantum is open (§11.7).
3. **Core-energy requirement.** Eq. (24) demands $f_{\text{core}} \approx 0.18\text{--}0.26$, above the rough $1/\ln(r_{\text{loop}}/\xi)$ estimate of Section 4.1; the GPE calculation decides. The target moves if the vortex-energy bracket constant ($-\frac{1}{2}$) is revised against the literature.
4. **Synchronization.** Universal phase-locking of the pumps (in-phase for like signs) is taken from the injection-locking behavior of mutually coupled self-sustained oscillators and is not derived here.
5. **Isolated-charge energetics.** The reactive (standing) character of a single charge's spin-phase field, as opposed to radiative loss at Ω , requires an explicit treatment.
6. **Spin-sector mode bookkeeping.** The statics above use only the kinetic stiffness (17), but photons-at- c requires the magnon branch at c while the cosmological spin-nematic modes of Paper 6 are slow; the mode inventory should be made explicit.

11. Structural Confirmation: The Bohr Magnetron from Ring Geometry

The derivation of §10 fixes the Coulomb constant k from the bound core ring's spin-phase source amplitude. A complementary, parameter-free check on the ring geometry comes from computing its classical magnetic moment and comparing to the observed Bohr magneton $\mu_B = e\hbar/(2m_e)$. The match is exact: no fit, no closure assumptions, no source-identification factor.

11.1 Effective Current of the Bound Core Ring

The ring carries charge e as a topological winding number (§9). The winding *pattern* advances azimuthally around the loop once per Compton period $T_C = 2\pi\hbar/(m_e c^2)$. This advance is a phase motion, not a material flow: the pattern speed at the loop radius is $\omega_C r_{\text{loop}} = c$ exactly, while the host circulation at that radius moves constituents at only $v(r_{\text{loop}}) = m_e c/(2m_A) \approx 4 \times 10^{-4} c$. (Were the advance material transport, the combined azimuthal and poloidal speeds of the core constituents would exceed c ; the winding advance is the motion of a pattern in the relative phase, not of the constituents that carry it.) An external observer averaging over the loop therefore sees one quantum of charge pass any azimuthal cross-section per Compton period:

$$I_{\text{eff}} = \frac{e}{T_C} = \frac{e \omega_C}{2\pi} = \frac{e m_e c^2}{2\pi \hbar}. \quad (25)$$

11.2 Two Windings, Two Clocks

The azimuthal advance above is one of *two* rotations the bound structure carries, and they must not be conflated. The same relative-phase winding also rotates poloidally—about the core tube’s own axis—at the rate inherited from the host vortex flow at the formation boundary, $\Omega = v(\xi)/\xi = m_A c^2/\hbar$ (Eq. 19): a property of the medium, common to every singly-wound structure regardless of its loop radius. The two clocks differ by $\Omega/\omega_C = m_A/m_e \approx 1.3 \times 10^3$: the winding re-threads poloidally some thirteen hundred times per azimuthal cycle—a fine-pitch screw. The division of labor is clean. The poloidal rotation is material (the bound core co-rotates with the flow that created it, at $v(\xi) = c/\sqrt{2}$, safely subluminal) and is the clock of the Coulomb source (§10.4). The azimuthal advance is pattern motion and is the clock of the magnetic moment computed here. The poloidal rotation is also compatible with everything below: its local rotation axes point azimuthally and close around the loop, so it contributes no net angular momentum about the symmetry axis and leaves both μ_M and S untouched.

11.3 Magnetic Moment of the Loop

Treating the ring as a closed current loop of radius r_{loop} and area $\mathcal{A} = \pi r_{\text{loop}}^2$, the classical magnetic moment is $\mu_M = I_{\text{eff}} \mathcal{A}$. Substituting $r_{\text{loop}} = \hbar/(m_e c)$ from Paper 7:

$$\mu_M = \frac{e m_e c^2}{2\pi \hbar} \cdot \pi \cdot \frac{\hbar^2}{m_e^2 c^2} = \frac{e \hbar}{2m_e} = \mu_B. \quad (26)$$

The cancellation in (26) is exact and parameter-free.

11.4 What the Match Confirms

Equation (26) requires three independently-motivated structural inputs to combine in precisely the right way:

- The ring radius is the reduced Compton wavelength: $r_{\text{loop}} = \hbar/(m_e c)$. Any other length scale produces the wrong answer.
- The azimuthal clock is the Compton frequency: $\omega_C = m_e c^2/\hbar$. This is the frequency of the winding pattern's advance around the loop (Paper 7), distinct from the poloidal clock Ω of §11.2.
- The effective current is one charge per period: $I_{\text{eff}} = e \omega_C/(2\pi)$. This follows from the topological character of charge established in §9: the ring carries a single winding whose azimuthal pattern advances exactly once per cycle.

Each input is motivated independently elsewhere in the framework, and their product yielding exactly μ_B is a non-trivial geometric self-consistency. As developed below, the third input is not unique to this calculation: the same one-winding-per-cycle normalization independently fixes the Coulomb source amplitude, making the pair of results a single-postulate structure rather than a coincidence.

11.5 Gyromagnetic Ratio $g = 2$

The same construction gives the Dirac value of the gyromagnetic ratio without further assumptions. The bound core ring carries quantized half-circulation $\kappa_e = \hbar/(2m_A)$ (cf. Paper 7 [6]), which corresponds to spin angular momentum $S = \hbar/2$. Combining with (26):

$$\mu_B = g \cdot \frac{e}{2m_e} \cdot S \implies g = 2, \quad (27)$$

to leading order. (The identification is exact per constituent—each participant in the half-quantum flow carries $m_A \kappa_e / 2\pi = \hbar/2$ —while the bookkeeping that assigns the *structure* a total of exactly $\hbar/2$ is carried by Paper 7's quantized-circulation argument and is not re-derived here.) The standard QED anomalous corrections $g = 2 + \alpha/\pi + O(\alpha^2)$ are expected to arise from coupled fluctuations of the surrounding Aether condensate (the analog of the radiative diagrams in QED) and lie beyond the classical ring picture.

11.6 What This Does *Not* Determine

The match in (26) yields μ_B in SI units of $\text{A} \cdot \text{m}^2$ from purely geometric and kinematic ring properties. It does *not* fix the vacuum permeability μ_0 . That constant enters only when one computes the

magnetic field produced by the moment at distance r ($B = \mu_0 \mu_M / (4\pi r^3)$) or the interaction energy of two such moments, and it is fixed ($\mu_0 = 4\pi k / c^2$) by the Coulomb-constant derivation of §10, whose source amplitude is normalized by the *same* postulate used in Eq. (25): one winding advance per Compton cycle, giving pump strength $q_0 = 2\pi r_{\text{loop}}$ —one winding quantum across the ring aperture per cycle. The two calculations therefore share their single dimensionless input:

- one winding per cycle, counted as an azimuthal current $\rightarrow \mu_B$, exactly;
- one winding per cycle, counted as a spin-phase pump $\rightarrow k$, to 0.2% at the reference point of §10.5.

One topological postulate, two observables. This is the sense in which the product in the preceding subsection is quantized rather than tuned: a normalization adjusted by hand to fit k would have no reason to land the magneton exactly, and vice versa.

11.7 Species Scaling: Why μ Scales and e Must Not

The construction extends across the charged leptons by $r_{\text{loop}} \rightarrow \hbar / (mc)$ and $T_C \rightarrow 2\pi\hbar / (mc^2)$, predicting $\mu = e\hbar / (2m)$ —as observed: the muon’s moment is $e\hbar / (2m_\mu)$ with the same $g \approx 2$. The scaling succeeds precisely because e enters (25) as a universal input while only the geometry scales with the species mass. This is a structural constraint on the companion Coulomb derivation: the charge quantum must *not* inherit the r_{loop} scaling that the moment correctly does—a source strength proportional to each species’ own loop radius would make the muon’s charge smaller than the electron’s by m_e / m_μ , against observation. The resolution direction is available within the framework: m_e is itself fixed by the medium through the self-consistent mass equation (§4.3), so $q_0 = 2\pi\hbar / (m_e c)$ is a constant of the medium—the charge quantum of the condensate’s ground ring mode—to be inherited by any charged structure, whatever its own size and clocks. Making that inheritance mechanism explicit is an open problem (§12).

11.8 Status

Derived (D). Given (i) the ring radius from Paper 7, (ii) the topological character of charge (§9), and (iii) the half-quantum circulation from Paper 7, the Bohr magneton and $g = 2$ follow with no free parameters or fits. The shared one-winding-per-cycle normalization ties this result to the Coulomb source amplitude, elevating the two to a single-postulate pair. **Open:** the total angular-momentum bookkeeping behind $S = \hbar/2$, and the species-inheritance mechanism for the charge quantum (§11.7).

12. Open Questions

1. **DCE permanence threshold.** What is v_{perm} ? A first-principles DCE calculation in vortex geometry would determine the equilibrium core size and the KE/ring mass partitioning. This is the key open calculation needed to make the model fully quantitative.
2. **Core energy in the logarithmic potential.** Computing the gradient and potential energy of a vortex core in the logarithmic condensate would determine f_{core} and, combined with the mass equation, pin down m_A uniquely. The Coulomb constraint makes this fully quantitative: joint consistency requires $f_{\text{core}} \approx 0.18\text{--}0.26$ (Eq. 24), a pre-registered target for the calculation.
3. **Muon and tau.** Could higher-energy vortex configurations (multiply-wound cores, knotted rings) stabilize at different masses through the same self-regulation mechanism? This is Paper 7 [6], Open Question 11.
4. **Form factor—residual geometric corrections.** See Section 9 for the full analysis. The revised Paper 4 [4] identifies charge as the topological rotation coupling between ring and vortex, which yields $F(Q^2) = 1$ exactly at the monopole level. A narrower remaining question is whether subleading geometric corrections of order $(Qr_{\text{loop}})^2$ arise from the ring's finite extent in higher multipole moments of the coherent spin-phase wave field.
5. **Winding-pump closure.** Section 10 derives the Coulomb constant from the spin-phase winding pump with one normalization postulate ($q_0 = 2\pi r_{\text{loop}}$; one winding quantum per cycle, Eq. 20). Remaining first-principles work: (a) the boundary-layer conversion calculation—the driven two-component GP problem at the rotating winding boundary, with pre-registered answer: duty $2/\pi$ of the layer population per poloidal turn; (b) the species-inheritance mechanism for the charge quantum (§11.7); (c) universal synchronization of the pumps from injection locking; (d) the isolated-charge energetics of the spin-phase field.
6. **Cosmological formation.** What conditions in the early universe create vortex rings with the minimum circulation quantum? What determines the initial r_{loop} , and how does it relax to $\hbar/(m_e c)$?

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